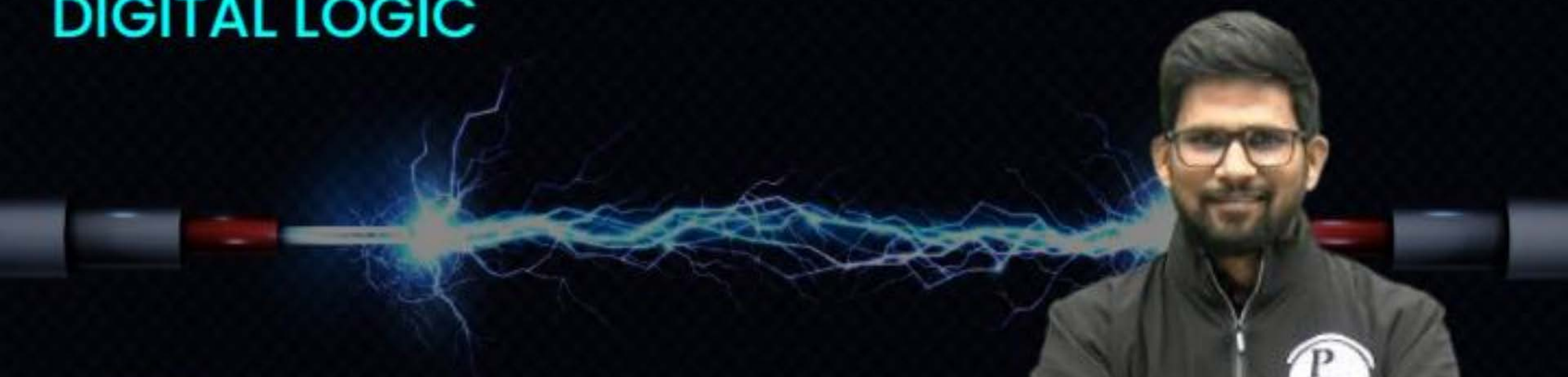
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 06

Combinational Circuit



By- Chandan Gupta Sir



F.S.

Basics of K-Map

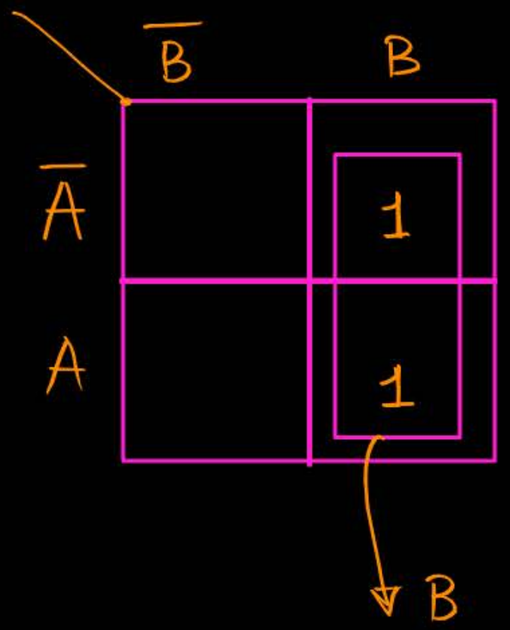


K-Map cont.

Question

$f(A, B) = \Sigma(1, 3)$

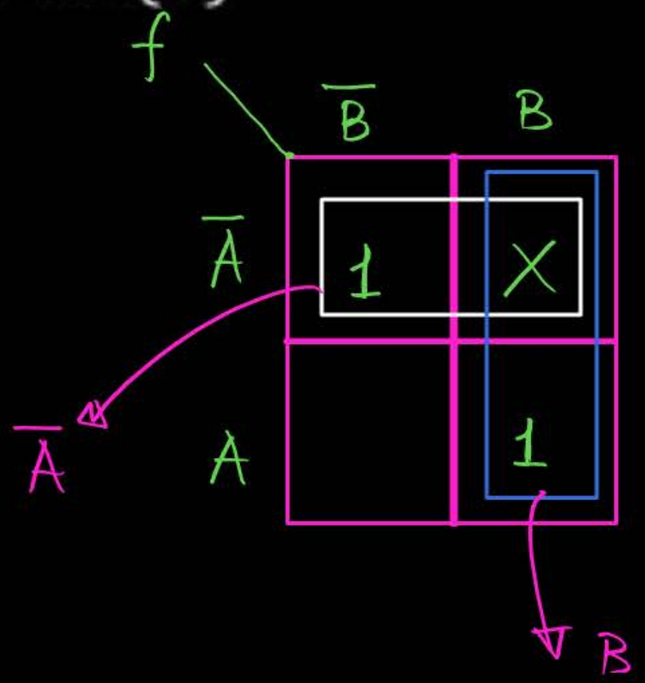
$= B$



Question

$f(A, B) = \Sigma(0, 3) + d\Sigma(1)$

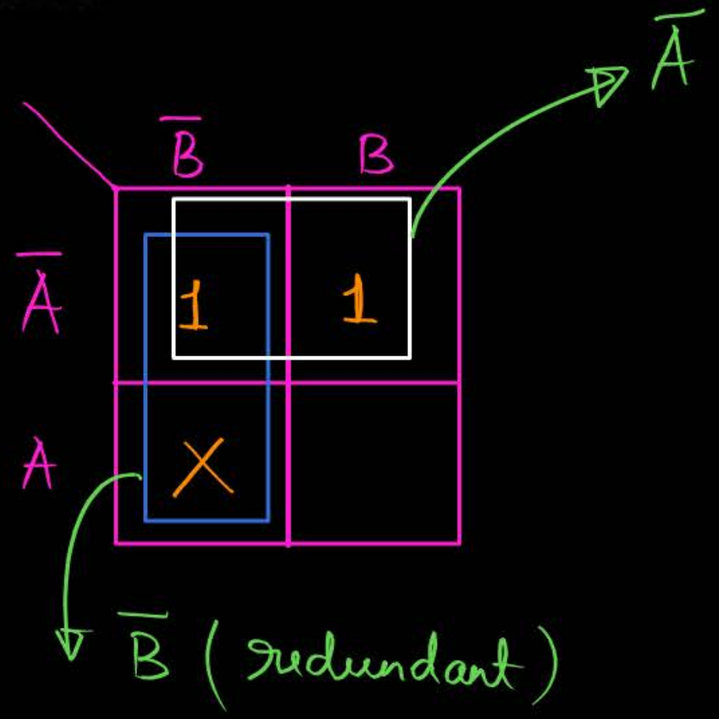
$= \overline{A} + B$

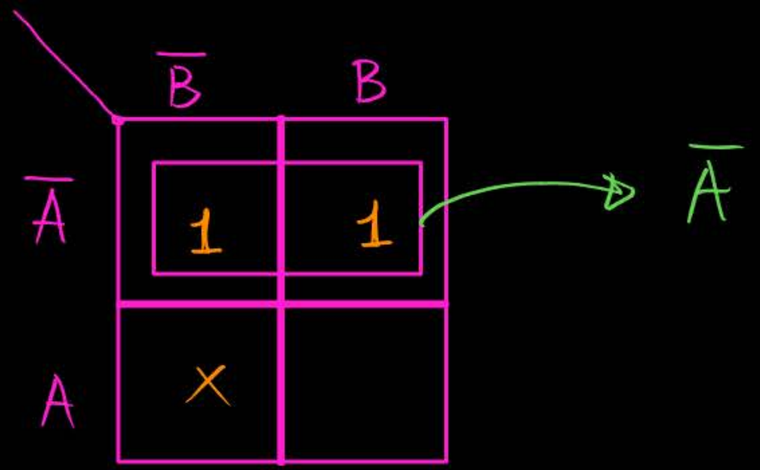


Question

$f(A, B) = \Sigma(0, 1) + d\Sigma 2$

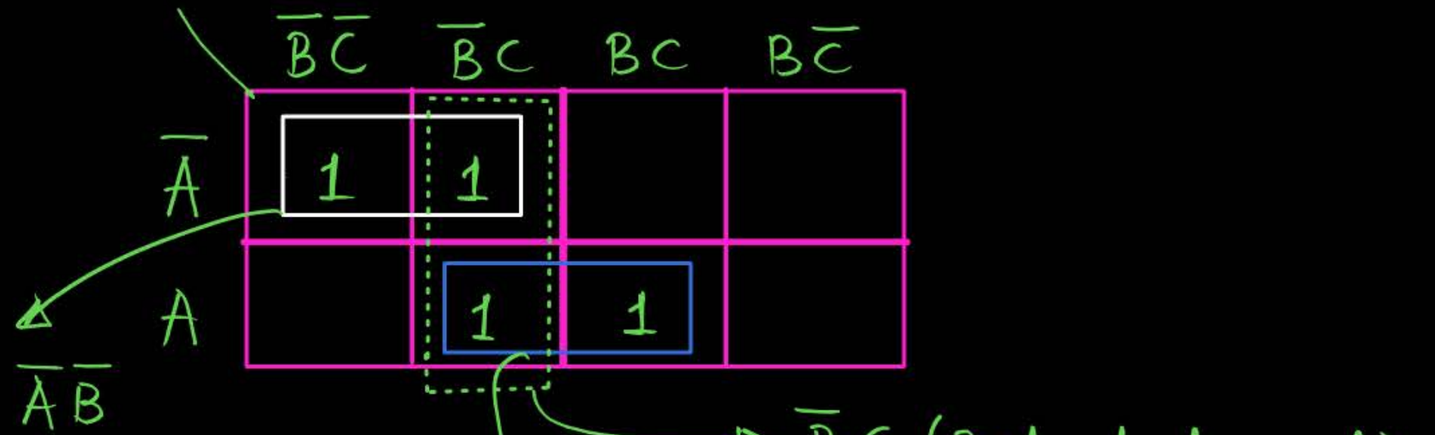
$= \overline{A}$





Question

$f(A, B, C) = \Sigma(0, 1, 5, 7)$

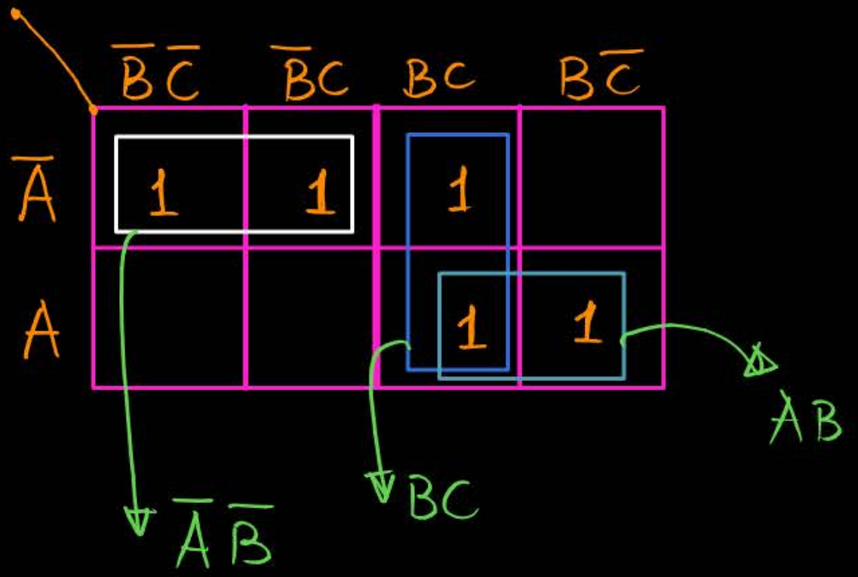


$= \bar{A}\bar{B} + AC + \bar{B}C$ $= \bar{A}\bar{B} + AC$

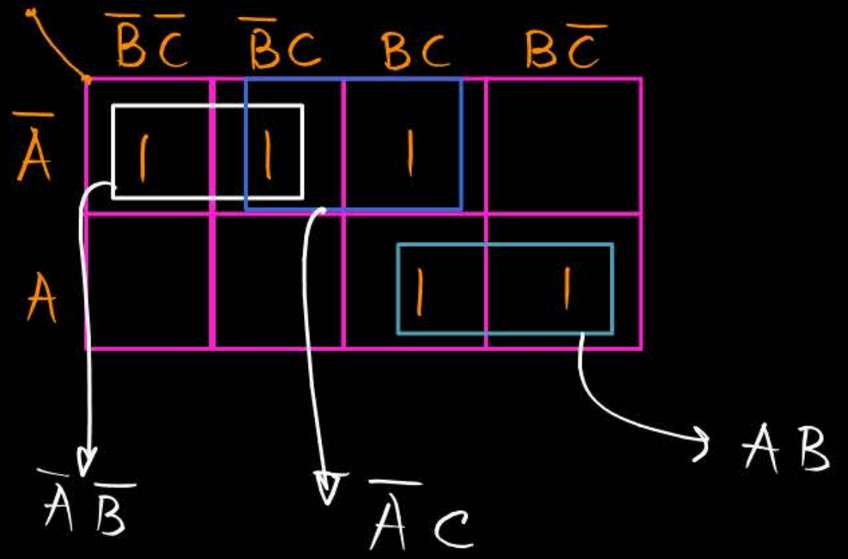
redundant term

Question

$f(A, B, C) = \Sigma(0, 1, 3, 6, 7)$



$= \bar{A}\bar{B} + AB + BC$

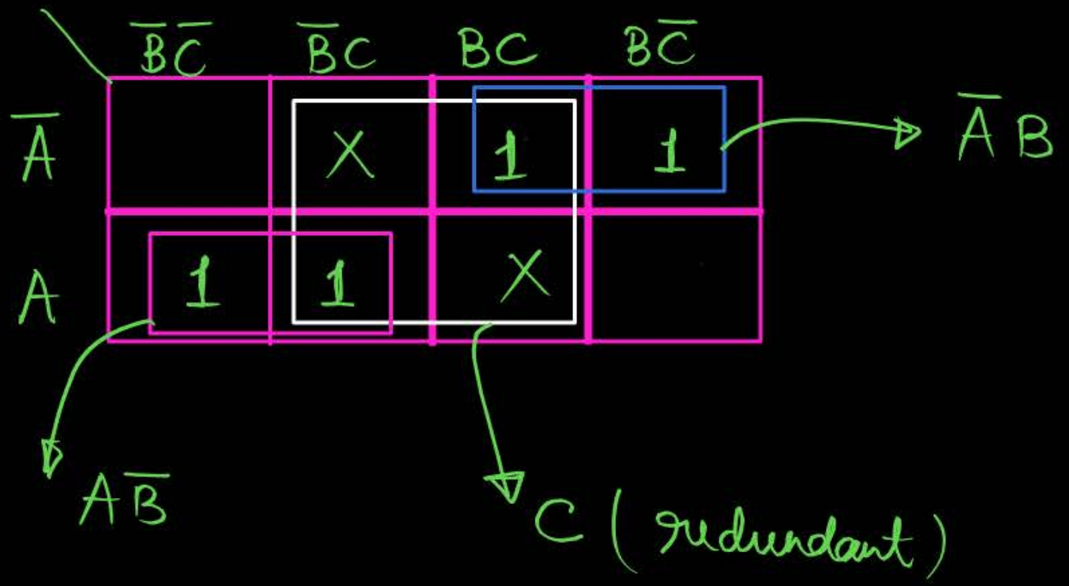


$= \bar{A}\bar{B} + AB + \bar{A}C$

Note: K-Map can have more than one solution but it is not always necessary.

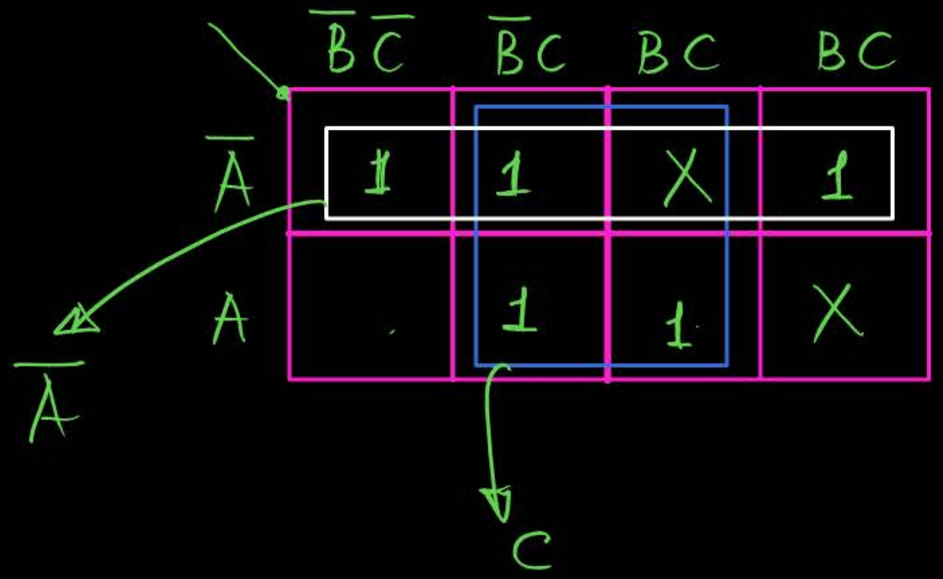
Question

$f(A, B, C) = \Sigma(2, 3, 4, 5) + d\Sigma(1, 7)$ $= A\overline{B} + \overline{A}B$



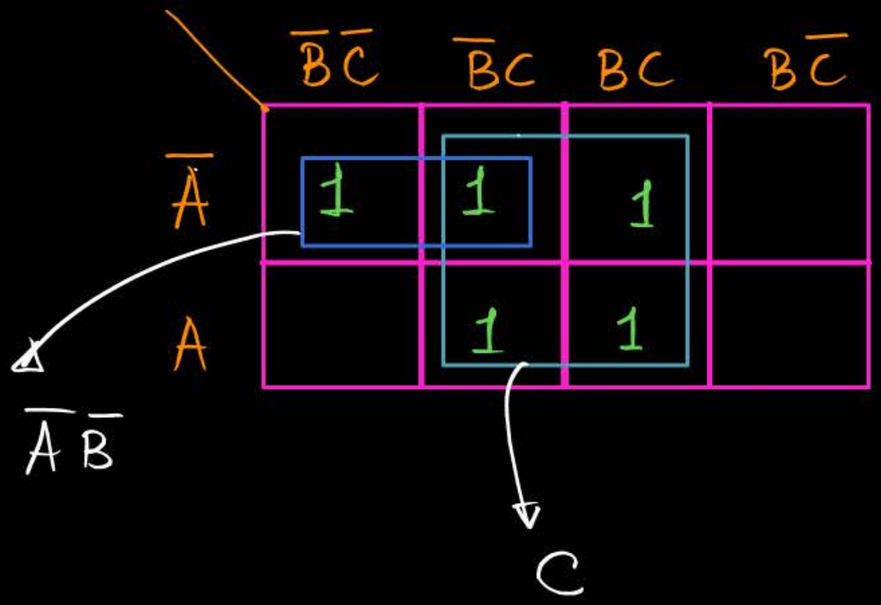
Question

$f(A, B, C) = \Sigma(0, 1, 2, 5, 7) + d\Sigma(3, 6)$ $= \overline{A} + C$



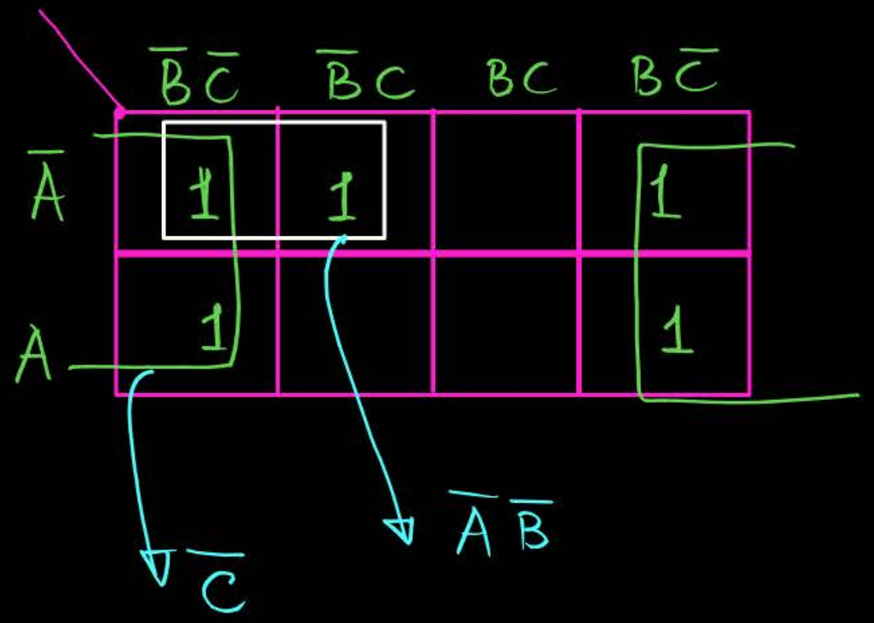
Question

$f(A, B, C) = \Sigma(0, 1, 3, 5, 7)$ $= (\bar{A}\bar{B} + C)$



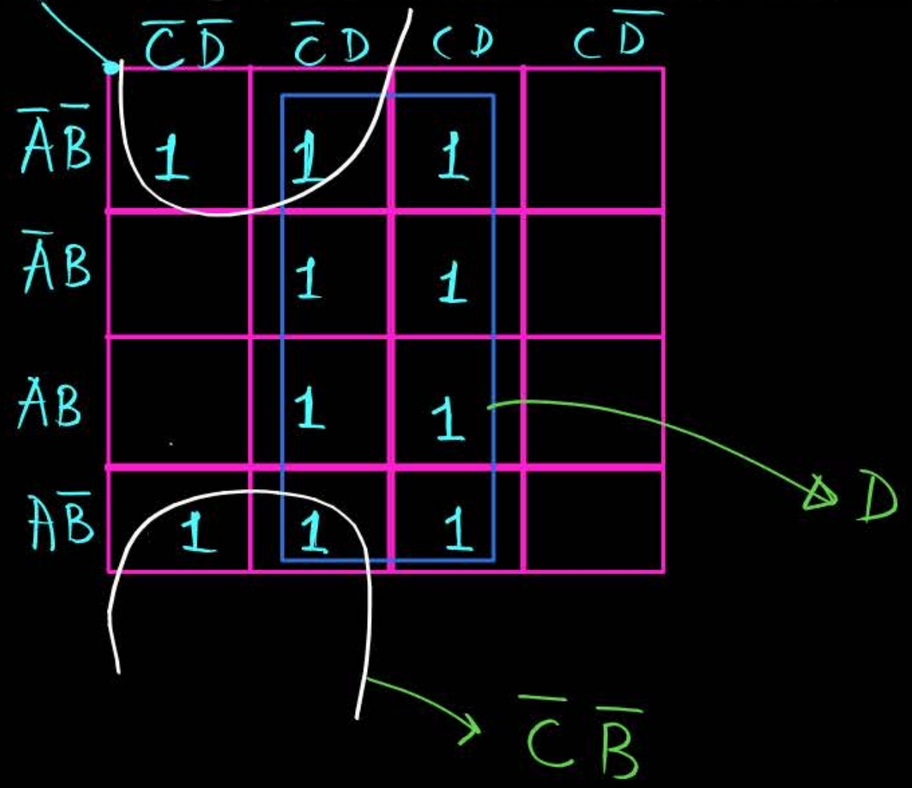
Question

$f(A, B, C) = \Sigma(0, 1, 2, 4, 6) = \bar{C} + \bar{A}\,\bar{B}$



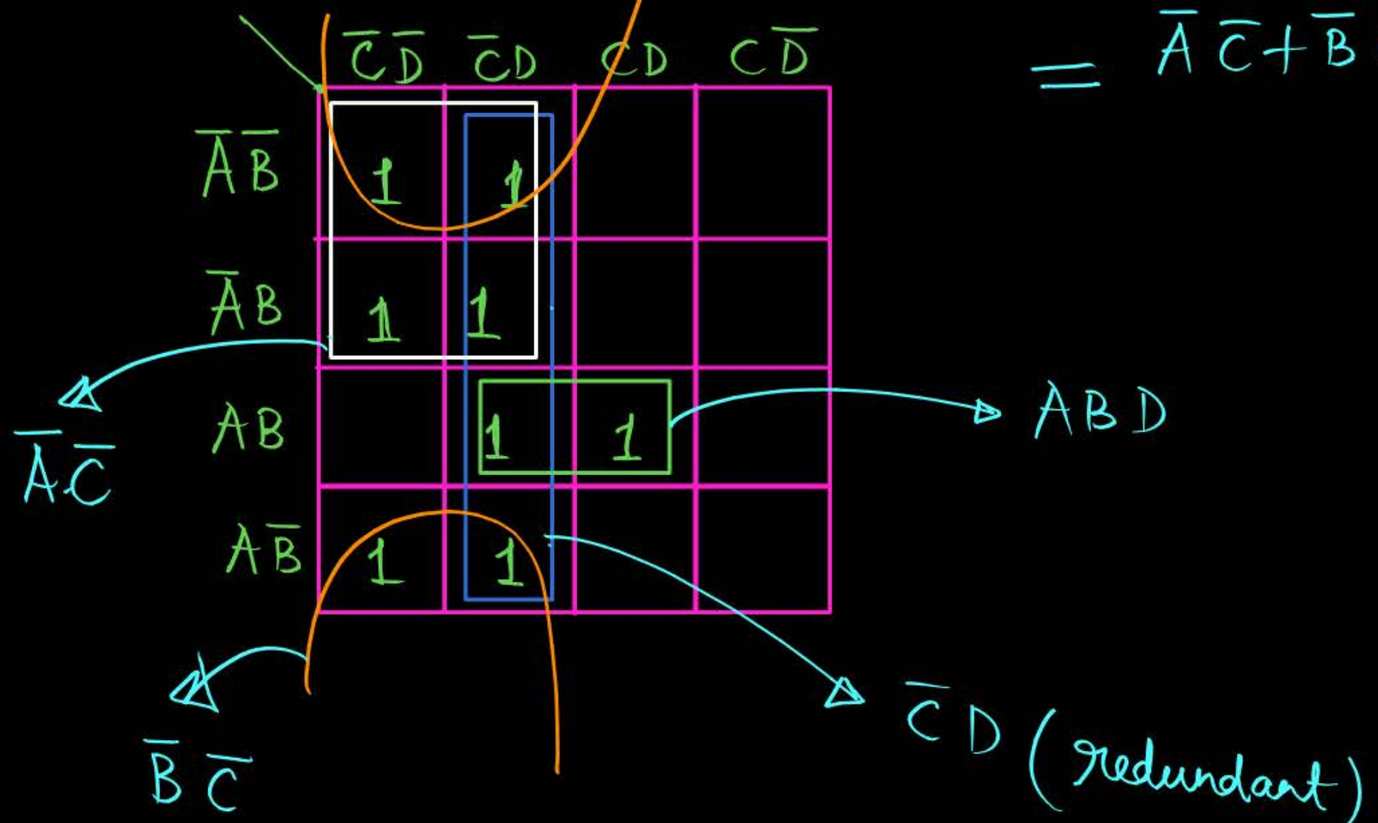
Question

$f(A, B, C, D) = \Sigma(0, 1, 3, 5, 7, 8, 9, 11, 13, 15) = \bar{B}\,\bar{C} + D$



Question

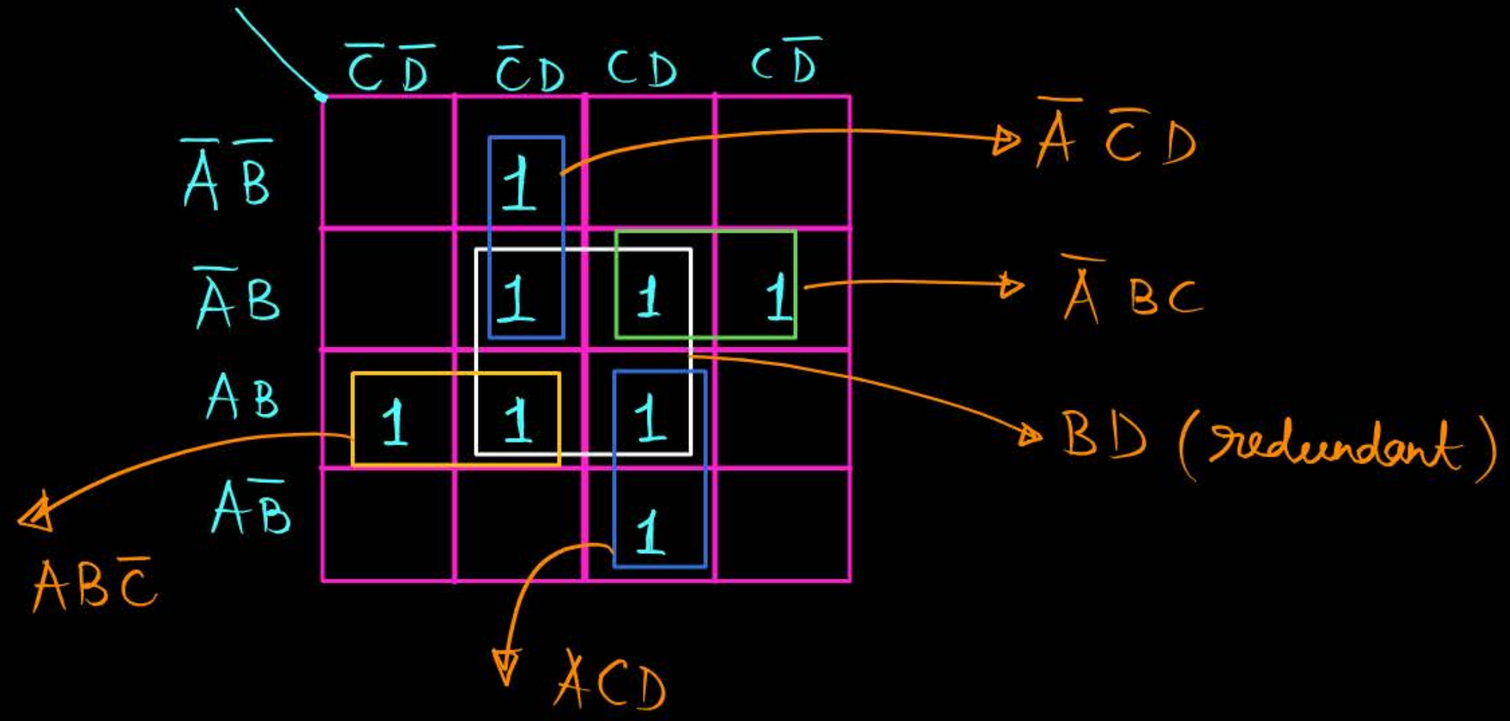
$f(A, B, C, D) = \Sigma(0, 1, 4, 5, 8, 9, 13, 15)$



$= \bar{A}\bar{C} + \bar{B}\bar{C} + ABD$

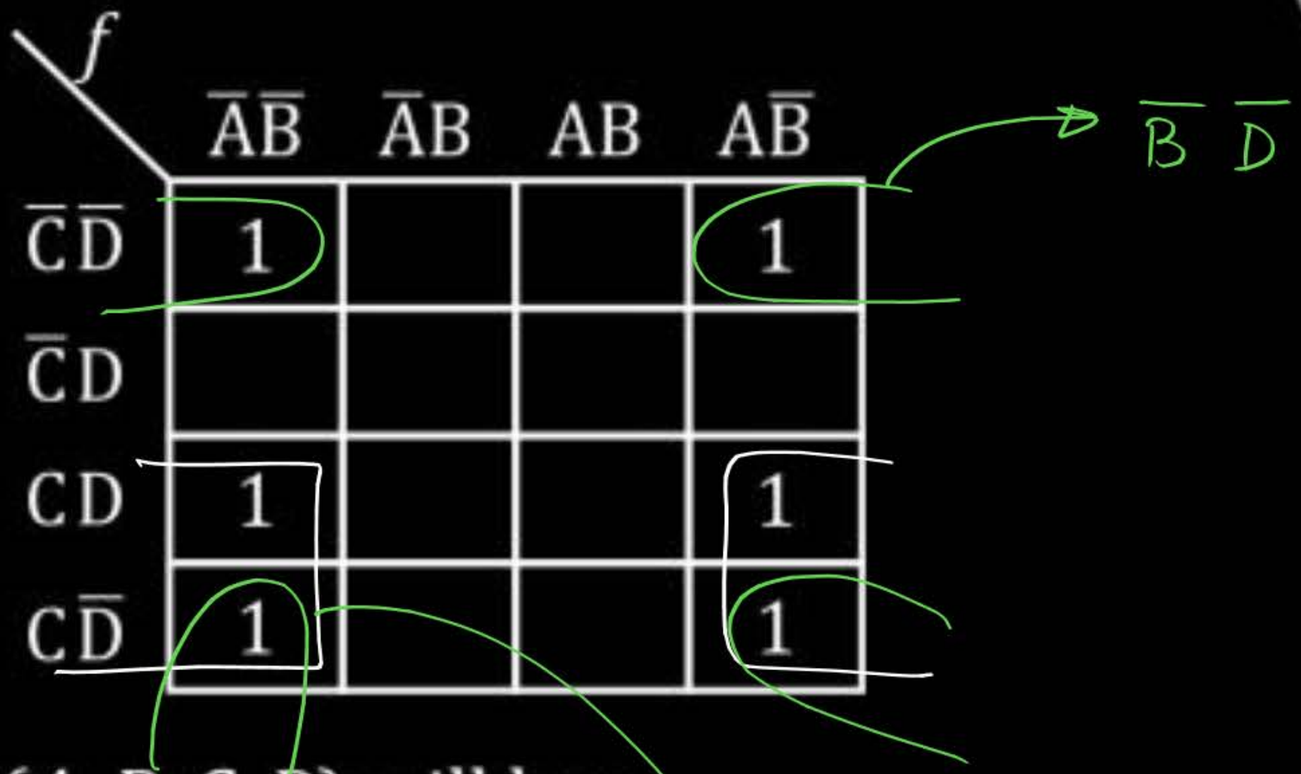
Question

$f(A, B, C, D) = \Sigma(1, 5, 6, 7, 11, 12, 13, 15)$ $= A\bar{B}\bar{C} + \bar{A}\bar{C}D + \bar{A}BC + ACD$



Question

A K-map is given as :



Minimized expression of $f(A, B, C, D)$ will be :

(a) $\bar{B}(C + \bar{D})$

(b) $\bar{A}\bar{B}C + A\bar{B}C + \bar{B}\bar{C}D$

(c) $\bar{A}\bar{B} + \bar{C}\bar{D}$

(d) None of these

$\bar{B}C + \bar{B}\bar{D}$

$= \bar{B}(C + \bar{D})$

Question

A K-map is given as :

f	$\bar{A}\bar{B}$	$\bar{A}B$	AB	$A\bar{B}$
$\bar{C}\bar{D}$	1	1		
$\bar{C}D$		X		
CD		X		
$C\bar{D}$	1	1	1	

H.W.

The minimized expression of above K-map is :

(a) $\bar{A}B + \bar{A}\,\bar{D} + BC\bar{D}$

(b) $\bar{A}\,\bar{D} + BC\bar{D}$

(c) $\bar{A}\bar{B}\bar{D} + \bar{A}B + BC\bar{D}$

(d) None of these

#Q.1.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1		
$\bar{A}B$	1	1	X	1
AB	1	X		1
$A\bar{B}$		X		

#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1		
$\bar{A}B$		1	X	
AB	1	X	X	1
$A\bar{B}$	1	1		

#Q.

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	
$\bar{A}B$		X	1	
AB		X	X	
$A\bar{B}$	1	1		



2 Minute Summary

K-Map

Thank you

GW

Soldiers !

